

CROSSSAFE · CALCULATION

CrossSafe sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	N1 recorded as decided on 2026-10-02 (existing-pole kit 76 mm and up until a slip-torque test); text only, no number changed
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.3	2026-09-30	Amish Chadha	Re-run for the constructable design (CRS-DDR-003); enclosure mounting plate in the wind area, PIR height, BOM of 23 lines; R15 now not met on paper
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); sun-shielded heat case, anti-rotation bolt and keyed saddles, shielded enclosure wind area, cost
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (flash timing, energy budget and autonomy, enclosure heat, radio link and latency, photometry screening, heights, wind load, clamps and footing, cost)

CrossSafe sizing calculations

On paper, CrossSafe meets ten of its fifteen requirements (four by calculation, six by design), has one at risk and one over its value-engineering target, and leaves three that only photometry, a site trial or a timed installation can settle. The one over target is cost (R15): the saddles, mounting plate, panel bracket and fixings that make the design buildable (CRS-DDR-003, 2026-09-30) take the estimated cost of one assembly \$10.00 over its \$350 value-engineering target. The fixes Amish accepted on 2026-09-25 (CRS-DDR-001 and CRS-DDR-002) do their job. Gating the radar with a PIR sensor, together with the IA-21 flash sequence (each head lit 25 % of the time, not the 35 % assumed at TRL 2), cuts the daily load from about 29.8 to 16.0 Wh and lifts autonomy from about 4.1 to 7.7 days (R6 met, 5.4 days at -20 °C). The ventilated sun shield cuts the enclosure's rise in full sun from 14.7 to 3.9 K when dusty, so the interior stays at 53.9 °C at 50 °C ambient and charging continues up to 41.1 °C ambient (R8 met on paper). The 114.3 x 3.6 mm post reaches 45 % of yield in a 40 m/s gust (R9 target 60 %). An eccentric gust on the sign twists it with 175 N·m; on new posts an M10 through-bolt carries this at 12 % of its bearing resistance, and on existing poles keyed saddles hold on poles of about 70 mm and larger, so R9 stays at risk only on smaller poles. The estimated cost of one assembly on an existing pole is \$360.00 against the \$350 value-engineering target, and of a two-sided crossing on existing poles \$720.00 against \$700 (R15 over the target by \$10.00 and \$20.00; cost drivers and savings are in the design decisions register). Every number in this note is printed by `docs/04-calcs/sizing.py` ; the tag in brackets, for example [B4], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that the device is safe to put beside a road. Photometry, detection, the battery in a sun-heated box, clamp preload and the foundation must be checked on hardware and by a qualified engineer, and the device needs road authority approval before any use on a public road. See CRS-PRC-001, Safety.

■ Scope and method

The note checks every requirement in CRS-REQ-001 v0.6 against the design in CRS-PRC-001 v0.6 and the parametric model `cad/src/model.py` . The script imports the model's `PARAMS` and `derived()` , so heights, wind areas, the post section and the footing are those of the STEP files and drawing CRS-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml` . Run it from the repo root with `python docs/04-calcs/sizing.py` ; it also writes `docs/04-calcs/results.csv` .

The design case is CRS-REQ-001's: a marked crossing on a two-lane road 7 m kerb to kerb, one assembly on each side, 300 activations of 20 s per day, 2.5 peak sun hours in the worst month and -20 to +50 °C ambient. Decisions referred to as D1 to D9 are in CRS-DDR-001; the sun shield and anti-rotation detail (E1, E2) are in CRS-DDR-002.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Flash pattern	800 ms sequence, 75 per minute; each indication lit 4 x 50 ms per sequence	FHWA IA-21, checked on the FHWA page on 2026-09-25
LED heads	6 W per lit head in daytime; drivers 91 % efficient; no credit taken for night dimming	As CRS-PRC-001 v0.2; to be measured
Pilot light	0.3 W, steady while flashing	Assumed
Standby	STM32WL-class radio in continuous LoRa receive 5.5 mA and controller plus PIR 0.1 mA, both at 3.3 V through an 80 % converter; 1 mA at 12.8 V for LED drivers off, fault indicator and BMS; charger self-consumption 4 mA at 12.8 V; radar 0.4 W when on	Typical class figures, assumed; the charger figure is written into BOM line 10
Radar gating	The PIR keeps the radar on 25 % of the day at a busy site	Assumed; sensitivity at 50 % and ungated
Solar	20 W panel; 2.5 peak sun hours worst month, 5 in a good month; panel heat and dust 0.85, MPPT 0.94, cell charge 0.95	As CRS-PRC-001 v0.2, now split into factors
Battery	12.8 V 12 Ah LiFePO ₄ , 80 % usable; 70 % capacity at -20 °C; 80 % at end of life; charge 0 to 45 °C; discharge to about 60 °C	Typical LiFePO ₄ data
Heat	Clear-sky beam 900 W/m ² , diffuse 10 % of beam, sun at 70° elevation on the worst azimuth; combined convection and radiation 10 W/m ² K; absorptance 0.45 clean light grey, 0.70 dusty; a ventilated shield passes 25 % of the solar gain; no credit for panel shade	Handbook ranges; the shield factor matches WWT-CAL-001 and is not measured
Radio	8-byte packet, SF7, 125 kHz, coding rate 4/5, 8-symbol preamble, explicit header; +14 dBm; 2 dBi antennas; 1 dB cable each end; 25 dB allowance for a bus between the assemblies; -123 dBm sensitivity; up to 2 retries; 868 MHz band with 1 % duty cycle	LoRa time-on-air formula; screening values
Photometry	Amber LED 50 lm/W; optic 80 %; beam 20° wide by 10° high	Assumed, for screening only
Wind	40 m/s gust; air 1.225 kg/m ³ ; drag coefficient 1.2 on every part including the post; S235 post, yield 235 MPa	As CRS-REQ-001; not a code check
Clamps	Band tension 1,000 N; friction 0.2; torsion from a quarter-width eccentricity of the sign pressure	Tension assumed, to be measured; eccentricity is a common allowance for uneven gust pressure on signboards
Anti-rotation	M10 A4-70 through-bolt, stress area 58 mm ² , shear resistance 0.6 fub As / 1.25; post wall bearing 2.5 α fu d t / 1.25 with α = 0.5 and fu = 360 MPa; keyed saddle grip friction 0.4	EN 1993-1-8 forms used as a screening check; the grip friction is assumed, to be measured
Footing	Non-constrained pole formula after IBC section 1807.3.2.1; lateral soil bearing 100 psf per ft of depth (clay), not doubled	Screening method; a local geotechnical and structural check governs

■ A. Flash time and pattern (R1, R4)

- **Time.** A 7 m crossing at 0.9 m/s takes 7.8 s; with 5 s added the set time is 12.8 s. The 20 s energy case covers crossings up to 13.5 m [A1].

- **Pattern.** IA-21 lights each indication for 200 ms of every 800 ms sequence, a duty of 25 %, not the 35 % assumed at TRL 2 [A2]. This alone cuts LED energy by 29 %.
- **Geometry.** Lenses of 140 x 62 mm exceed the 127 x 51 mm minimum; the bar bottom is at 2,100 mm and the sign's bottom point at 2,250 mm, so the bar sits directly under the sign as IA-21 allows [A3].

■ B. Energy, autonomy and recovery (R6, R7)

- **Loads.** The lights flash 1.67 h a day. Four heads draw 6.59 W on average while flashing, 10.99 Wh a day; the pilot light adds 0.50 Wh and radio transmission 0.001 Wh [B1].
- **Standby.** Logic 23.1 mW, 12 V quiescent 12.8 mW, charger 51.2 mW and the gated radar 100 mW average: 187 mW, or 4.49 Wh a day [B2]. At TRL 2 standby was a lumped 0.6 W, almost half the load.
- **Daily load.** 16.0 Wh a day, against the TRL 2 estimate of 29.8 Wh [B3].
- **Autonomy.** 122.9 Wh usable gives 7.69 days; 5.38 days at -20 °C and 6.15 days at end of life. The 18 Ah fallback (not adopted) would give 11.5 days [B4].

Table 2. Autonomy sensitivity [B5].

Case	Standby	Load	Autonomy
Baseline (PIR-gated radar, 25 % on)	187 mW	16.0 Wh/day	7.69 days
Radar on 50 % of the day	287 mW	18.4 Wh/day	6.69 days
TRL 2 controller (0.15 W) and a 10 mA charger, gated	391 mW	20.9 Wh/day	5.89 days
Radar ungated (no PIR)	487 mW	23.2 Wh/day	5.30 days
TRL 2 lumped standby (0.6 W)	600 mW	25.9 Wh/day	4.75 days

R6 holds in every case except the TRL 2 lump, which the build-up above replaces. The margin rests on a low self-consumption charger and on the PIR gating, both of which are in the BOM.

- **Harvest.** The chain from panel rating to stored energy is 75.9 % efficient. In the worst month the panel stores 38.0 Wh a day, a margin of 22.0 Wh. Break-even is 1.05 peak sun hours. At 5 peak sun hours the surplus is 59.9 Wh a day, which refills the battery from 20 % in 2.05 days [B6].
- **Currents.** Peak charge 1.25 A (0.10 C); average discharge while flashing 0.58 A, and 2.06 A in the 50 ms when all four heads are lit, so the 5 A battery fuse has margin [B7].

■ C. Enclosure heat and the charging window (R8)

The 150 x 260 x 300 mm enclosure has 0.324 m² of surface and presents 0.0674 m² to a 70° sun on its worst azimuth [C2].

Table 3. Enclosure temperature in full sun [C1, C3]. The dusty, shielded case is the design case (CRS-DDR-002).

Case	Rise over ambient	Interior at 45 °C	Interior at 50 °C	Charging stops above
Clean light grey	9.6 K	54.6 °C	59.6 °C	35.4 °C ambient
Dusty	14.7 K	59.7 °C	64.7 °C	30.3 °C ambient
Dusty, with the ventilated shield (design)	3.9 K	48.9 °C	53.9 °C	41.1 °C ambient

The BMS and charger stop charging above 45 °C cell temperature as R8 requires. Without a shield the device would lose harvest on hot, clear days above about 30 °C ambient, and at 50 °C ambient a dusty enclosure would reach 64.7 °C, above the typical 60 °C discharge limit of LiFePO4 cells. Amish accepted the recommended fix on 2026-09-25 (CRS-DDR-002): a ventilated white sun shield (part 18, about \$8, as in WaterWatch and FieldNode) over the roof and three walls with a 25 mm air gap. With it the rise is 3.9 K, the interior 53.9 °C at 50 °C ambient and charging continues up to 41.1 °C ambient [C3], so R8 is met on paper. The steady-state model ignores the battery's thermal mass, which lags the peak, and gives no credit for shade from the panel 0.78 m above the enclosure top; the shield factor of 0.25 is borrowed from WWT-CAL-001 and not measured. Moving the enclosure under the panel's shadow is to be considered when the bracket is detailed.

■ D. Radio link and latency (R3)

- **Time on air.** An 8-byte packet at SF7 and 125 kHz takes 36.1 ms [D1].
- **Latency.** Wake, transmit, receive and switch on the far side take 48 ms; with two retries after lost acknowledgements, 244 ms, inside the 500 ms target [D2].
- **Link budget.** Across 7 m with 25 dB of blockage from a bus, the far side receives -57.1 dBm, 65.9 dB above sensitivity; across 15 m, -63.7 dBm and 59.3 dB [D3]. The link will not limit the design; interference and regional band rules are the real questions.
- **Duty cycle.** Sixty activations in the busiest hour use 4.3 s of airtime per side against 36 s allowed at 1 % [D4].

■ E. Photometry screening (R2)

A 6 W amber head with the assumed efficacy and a 20° x 10° beam gives about 3,939 cd on axis: 0.39 lx at a driver's eye at 100 m and 4.38 lx at 30 m. At 100 m the near-lane driver is 2.0° off axis horizontally and 0.6° below the head; at 30 m, 6.7° and 2.0°, both inside the beam [E1]. IA-21 requires the daytime intensity to meet the SAE J595 Class 1 yellow peak luminous intensity; that standard's figures were not available to check, so R2 is not verifiable at TRL 3. The numbers above are the input a photometric check will need, not a result.

■ F. Heights, mounting and installation (R10, R11, R14)

- **Heights.** Button center 1,050 mm (R11: 0.9 to 1.1 m); light bar bottom 2,100 mm; enclosure bottom 2,850 mm (R14: 2.8 m), with the shield open at the bottom; radar at 2,600 mm on its arm with the PIR underside at 2,515 mm, so no part or cable of the sensing head is below 2.5 m; overall height 4,031 mm [F1].
- **Poles.** The band clamps fit 60 to 114.3 mm poles; the new-post variant uses 114.3 x 3.6 mm [F2].
- **Installation.** A task estimate for a two-person crew on an existing pole comes to 120 min, exactly the 2 h limit of R10, with the sun shield fitted to the enclosure on the ground and the keyed saddles in place of plain ones [F3]. Only a timed trial can settle it; Amish accepted that trial on 2026-09-25, and it is on hold with TRL 4.

■ G. Wind, clamps and footing (R9)

- **Loads.** The dynamic pressure at 40 m/s is 980 Pa. Wind along the road, on the sign face, gives 1,448 N and a base moment of 3,521 N·m (sign 1,839, post 920, shielded enclosure with its mounting plate 449, light bar 238 N·m) [G1]. The shield, and the mounting plate's edges and saddle tabs beside it (CRS-DDR-003), enlarge the enclosure's wind area. Wind across the road gives 759 N and 1,644 N·m, of which the panel is 416 N·m [G2]. Along the road governs.
- **Post.** *Table 4. Post stress in the 40 m/s gust [G3, G4].*

Post	Section modulus	Stress	Share of S235 yield	R9 (60 %)
76.1 x 3.2 mm	12,820 mm ³	251 MPa	107 %	Yields
88.9 x 4.0 mm	21,674 mm ³	153 MPa	65 %	Not met
88.9 x 4.0 mm with a 600 mm sign	21,674 mm ³	122 MPa	52 %	Met
114.3 x 3.6 mm (decided, D3)	33,593 mm ³	105 MPa	45 %	Met

The TRL 2 figures (169 MPa and 72 % on the 88.9 mm post) counted the panel face-on in the same gust as the sign; here the two act in different wind directions. The top of the 114.3 mm post deflects about 31 mm (0.8 % of its height) in the gust [G5].

- **Clamps.** A gust whose pressure centre sits a quarter of the sign's width off the pole axis twists the sign clamps with 175 N·m. Two bands at the assumed 1,000 N tension resist about 144 N·m by friction on a 114.3 mm pole and only 75 N·m on a 60 mm pole [G6]. With plain saddles the sign could rotate on its pole in a storm. Amish accepted the recommended anti-rotation detail on 2026-09-25 (CRS-DDR-002):
- **New posts:** an M10 through-bolt in place of the upper sign saddle's upper sign bolt (part 19), through the sign, the saddle and both post walls. The 175 N·m couple puts 1,535 N on each post wall: 8 % of the bolt's shear resistance and 12 % of the wall's bearing resistance [G6b]. Met.
- **Existing poles:** keyed saddles with serrated grip faces (part 20). At an assumed grip friction of 0.4 two bands resist 287 N·m on a 114.3 mm pole and 151 N·m on a 60 mm pole, so the sign holds on poles of about 70 mm and larger [G6c]. Smaller poles remain at risk, and the friction and band tension must be measured (TRL 4, on hold). Decided 2026-10-02 (CRS-DDR-002, N1): the existing-pole kit is limited to poles of 76 mm and larger until a slip-torque test confirms the grip, then a third saddle and band brings 60 to 75 mm poles back. The R10 row below and `results.csv` still state the 60 to 114.3 mm clamp range until `sizing.py` is updated.
- **Footing.** The resultant of 1,448 N (326 lbf) acts 2.43 m (8.0 ft) above the sidewalk. The non-constrained pole formula gives 1,791 mm of embedment for a 500 mm footing in clay, so the model uses 1,800 mm [G7]. This is a screening size; the local code and soil govern.
- **Existing poles.** An assembly adds 2,601 N·m at the sidewalk to the host pole [G8]. The pole owner must confirm the pole and its foundation can carry it; that check is outside this repo.

■ H. Cost (R15)

The BOM has 23 lines, all priced. One assembly on an existing pole costs \$360.00 and \$433.00 with its own post and anti-rotation bolt, footing concrete excluded [H1]. The design for construction (CRS-DDR-003) added \$28.00: eight keyed saddles in place of two (\$18.00 more), the enclosure mounting plate (\$8.00), the radar arm and battery strap (\$3.00), a fuller panel bracket (\$6.00 more) and more fixings (\$4.00 more), less the sign maker's brackets

(\$5.00) and the bought band brackets (\$6.00). Against the value-engineering target of \$350 (`budget_usd` , a hypothetical control target, not a limit) that is \$10.00 over on an existing pole and \$83.00 over with a new post [H2]. Under D1 the target covers the existing-pole assembly, so R15 is over the target on paper; cost drivers and savings worth trying are in the Value engineering section of `docs/06-design-decisions.md` . A two-sided crossing costs \$720.00 on existing poles (\$20.00 over R15's \$700) and \$866.00 with new posts: \$30.00 under and \$116.00 over the \$750 site-trial value-engineering target, which Amish accepted on 2026-09-25 for when a site trial starts (on hold with TRL 4) [H3].

■ **L. Results against every requirement**

Table 5. Requirement status at TRL 3 (also in `docs/04-calcs/results.csv`).

ID	Requirement	Value	Target	Status
R15	Affordable	\$360.00 per assembly; \$720.00 per crossing on existing poles	\$350; \$700 per crossing	Over the value-engineering target by \$10.00 (\$20.00 per crossing)
R9	Structure	45 % of yield on the 114.3 mm post; bolt 12 % of bearing; keyed saddles hold on poles of 70 mm and up	60 % of yield or less; no sign rotation	At risk (poles under about 70 mm)
R2	Visibility	About 3,939 cd per head (screening)	Seen at 100 m in sun; dimmed at night	Not verifiable at TRL 3
R5	Detection quality	PIR wakes the radar; 1 s dwell	5 % false, 2 % missed	Not verifiable at TRL 3
R10	Mounting	Fits 60 to 114.3 mm poles; bar at 2.1 m; about 120 min estimate	Fit; 2.1 m; 2 h	Not verifiable at TRL 3 (install time)
R3	Activation and sync	48 ms; 244 ms with two retries	0.5 s or less	Met on paper
R6	Autonomy	7.7 days (5.4 at -20 °C; 6.2 at end of life)	5 days or more	Met on paper
R7	Energy balance	Break-even 1.05 peak sun hours; recovery 2.1 days	Neutral at 2.5 h; 3 days or fewer	Met on paper
R8	Environment	53.9 °C inside at 50 °C ambient (dusty, shielded); charging to 41.1 °C ambient	IP65; -20 to +50 °C; charge 0 to 45 °C	Met on paper
R1	Light geometry and pattern	140 x 62 mm lenses; IA-21 pattern; bar under the sign	127 x 51 mm min.; IA-21	Met by design
R4	Flash duration	12.8 s for 7 m; 20 s energy case	L/0.9 + 5 s; 10 to 40 s	Met by design
R11	Accessibility	Button 1.05 m; piezo; tone, tactile arrow, pilot light	0.9 to 1.1 m; 22 N or less	Met by design
R12	Fail-safe	Timer-limited flashing; local flashing without link; fault log	Never continuous	Met by design
R13	Privacy	Presence and counts only	No images or audio	Met by design
R14	Theft and vandal resistance	Enclosure bottom 2.85 m; lowest sensing part 2.52 m	2.8 m; no cables below 2.5 m	Met by design

■ Checks against the TRL 2 figures

Quantity	TRL 2 (CRS-PRC-001 v0.2)	TRL 3 (this note)	Reason
Flash time, 7 m	About 13 s	12.8 s	Confirmed
LED energy	About 15.4 Wh/day	10.99 Wh/day	IA-21 duty 25 %, not 35 %
Standby	About 14.4 Wh/day	4.49 Wh/day	PIR gating (D2) and a component build-up
Daily load	About 29.8 Wh/day	16.0 Wh/day	As above, plus the pilot light (D8)
Autonomy	About 4.1 days	7.69 days	As above
Break-even	About 2.0 peak sun hours	1.05	Lower load
Recovery at 5 peak sun hours	About 2.7 days	2.05 days	Lower load
Latency	About 0.1 s	48 ms, 244 ms with retries	Confirmed
Base moment	About 3.7 kN·m	3.52 kN·m	Panel and sign loads in different wind directions; shielded enclosure and its mounting plate
88.9 x 4 mm post	169 MPa, 72 %	153 MPa, 65 %	As above; still not met
114.3 x 3.6 mm post	110 MPa, 47 %	105 MPa, 45 %	As above
Cost per assembly	\$361 own post; \$311 existing pole	\$433.00; \$360.00	PIR, pilot light, larger post, sun shield, saddles, bolt, parts added for construction (CRS-DDR-003)

CRS-PRC-001, CRS-REQ-001 and README.md have been brought in line with the TRL 3 column.